Uses of Index number

i. It is an important tool for the formulating decision and management policies.

ii. It helps in studying the trends and tendencies.

iii. It determines the inflation and deflation in an economy.

Construction of Index Number

There are two types in construction of index number.

(i) Unweighted Index Number

(ii) Weighted Index Number

We confine ourselves to Weighted Index Number.

9.2.2 Weighted Index Number

In general, all the commodities cannot be given equal importance, so we can assign weights to each commodity according to their importance and the index number computed from these weights are called as weighted index number. The weights can be production, consumption values. If 'w' is the weight attached to a commodity, then the price index is given by,

$$\text{Price Index } (P_{01}) = \frac{\sum p_1 w}{\sum p_0 w} \times 100,$$

Let us consider the following notations,

p_1 - current year price

p_0 - base year price

q_1 - current year quantity

q_0 - base year quantity

where suffix '0' represents base year and '1' represents current year.

Laspeyre's price index number

$$P_{01}^{L} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

Paasche's price index number

$$P_{01}^{P} = \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100$$

Fisher's price index number

$$P_{01}^{F} = \sqrt{P_{01}^{L} \times P_{01}^{P}}$$

$$P_{01}^{F} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \times 100$$

Note

To get exact Fisher's price index number, one should use formula method rather than using $P_{01}^{F} = \sqrt{P_{01}^{L} \times P_{01}^{P}}$.

In Laspeyre's price index number, the quantity of the base year is used as weight.

In Paasche's price index number, the quantity of the current year is used as weight.

DO YOU KNOW?

Fisher's price index number is the geometric mean between Laspeyre's and Paasche's price index number

Example 9.10

Calculate the Laspeyre's, Paasche's and Fisher's price index number for the following data. Interpret on the data.

Commodities	Price		Quantity	
	2000	2010	2000	2010
Rice	38	35	6	7
Wheat	12	18	7	10
Rent	10	15	10	15
Fuel	25	30	12	16
Miscellaneous	30	33	8	10

Solution

Commodities	Price		Quantity		$p_0 q_0$	$p_0 q_1$	$p_1 q_0$	$p_1 q_1$
	2000 (p_0)	2010 (p_1)	2000 (q_0)	2010 (q_1)				
Rice	38	35	6	7	228	266	210	245
Wheat	12	18	7	10	84	120	126	180
Rent	10	15	10	15	100	150	150	225
Fuel	25	30	12	16	300	400	360	480
Miscellaneous	30	33	8	10	240	300	264	330
				Total	**952**	**1236**	**1110**	**1460**

Table 9.9

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